

QuantumCylinder

Measurement basis as a control knob for projected quantum denoising

We compare random-unitary and Hamiltonian projected diffusion with shared fidelity-based metrics, then use Problem 3(b) trade-off analysis to motivate two-way post-selection in Problem 3(c).

Scope

Small 2-qubit/3-qubit state-vector benchmark with explicit limitations.

Core claim

Recoverability, success probability, and diversity must be read together.

Deliverables

Final notebook, figures, tables, source code, tests, and presentation material.

Problem Map and Judging Criteria

Problem 1

Target ensemble around $|00\rangle$, metrics, random-unitary diffusion, Haar-like reference plateau.

Problem 2

Fixed Hamiltonian projected diffusion and qualitative control-cost comparison.

Problem 3

Measurement-induced denoising, basis trade-off, and two-way post-selection.

Judging alignment

- Completeness: every subproblem 1(a) through 3(c) is explicitly answered.
- Appropriateness: claims are scoped to the benchmark and reported trade-offs.
- Novelty: measurement basis is treated as an effective non-unitary map control knob.
- Presentation: first 5 minutes cover the full story; appendix supports Q&A.

Problem 1: Random-Unitary Diffusion

- S_0 is a 2-qubit target ensemble clustered near $|00\rangle$.
- Fidelity, MMD, and Wasserstein-type cost $1-F$ define the shared metric layer.
- Haar reference: MMD 0.869583 ± 0.024043 ; Wasserstein 0.724439 ± 0.021491 .
- Interpretation: rapid strong scrambling toward a Haar-like high-distance plateau, not slow Gaussian DDPM-like diffusion.

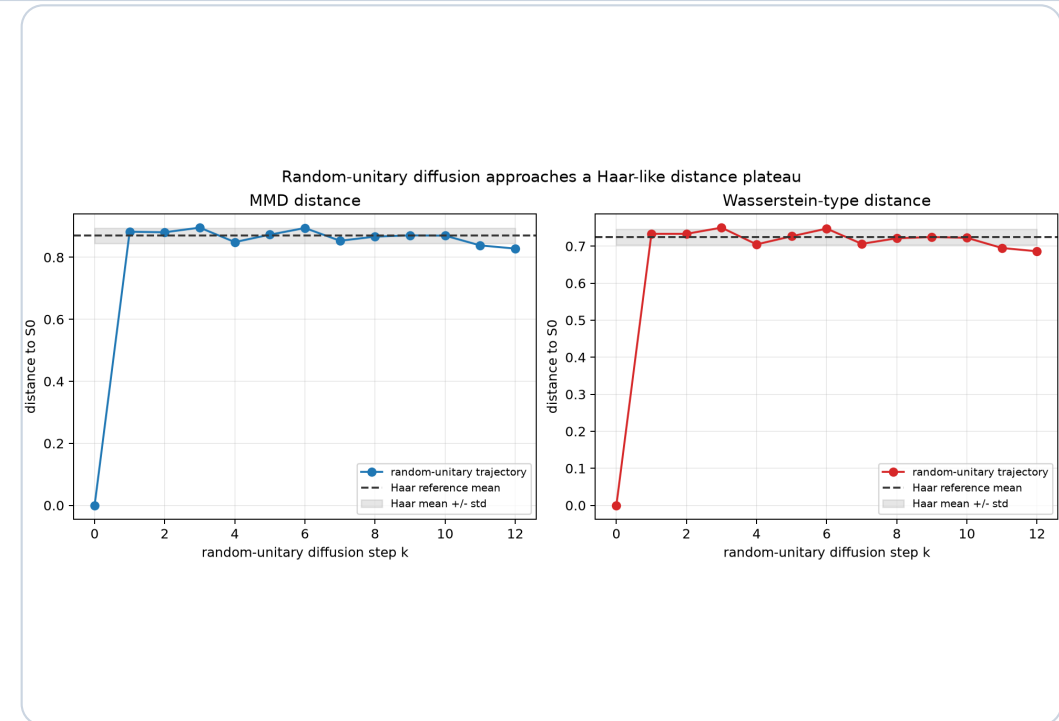


Figure: solution/figures/fig2_random_unitary_haar_baseline.png

Problem 2: Hamiltonian Projected Diffusion

- A 2-qubit data system M is extended with one complement qubit F .
- Fixed Hamiltonian evolution followed by projection constructs the projected ensemble.
- The Hamiltonian curve can fluctuate or partially saturate depending on time schedule and projection choice.
- Problem 2(d) compares resource/control proxies: random gate controls vs fixed Hamiltonian time and projection basis.

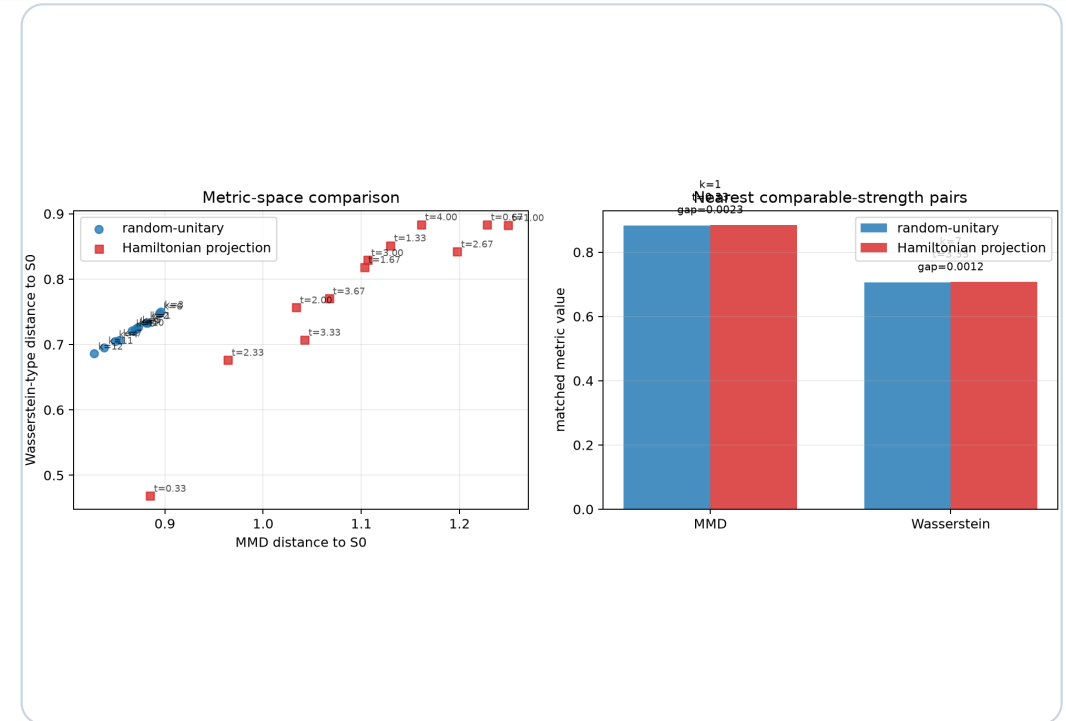


Figure: solution/figures/problem_1_2_metric_aligned_comparison.png

Problem 3(b): Measurement basis controls the recoverability-success-diversity trade-off

- M+F evolves unitarily, but measuring F and post-selecting induces an effective non-unitary map on M.
- Axis-only Z/X/Y is an interpretable Pauli-basis baseline.
- Continuous Bloch-sphere basis is a controlled generalization.
- The axis-only margin is small; the main result is the trade-off, not overwhelming continuous-basis superiority.

method	MMD	W	success	diversity
input	0.000000	0.000000	1.000000	1.000000
axis-only	0.086055	0.142594	0.465058	0.810592
continuous	0.097056	0.147983	0.468122	0.823217
collapse diagnostic	0.859292	0.714276	1.000000	0.000000

Interpretation column is traceable in `solution/tables/problem3b_measurement_basis_tradeoff.csv`

Problem 3(c): Analysis-guided improvement via two-way projected denoising

- Because 3-b shows distance gain can trade off against success probability, 3-c tests a stronger two-way Hamiltonian post-selection step.
- Continuous reference: MMD gain 0.056388; Wasserstein gain 0.120620; success 0.467554.
- Hamiltonian two-way: MMD gain 0.101374; Wasserstein gain 0.136426; success 0.227065.
- Conclusion: stronger distance improvement with lower post-selection success probability.

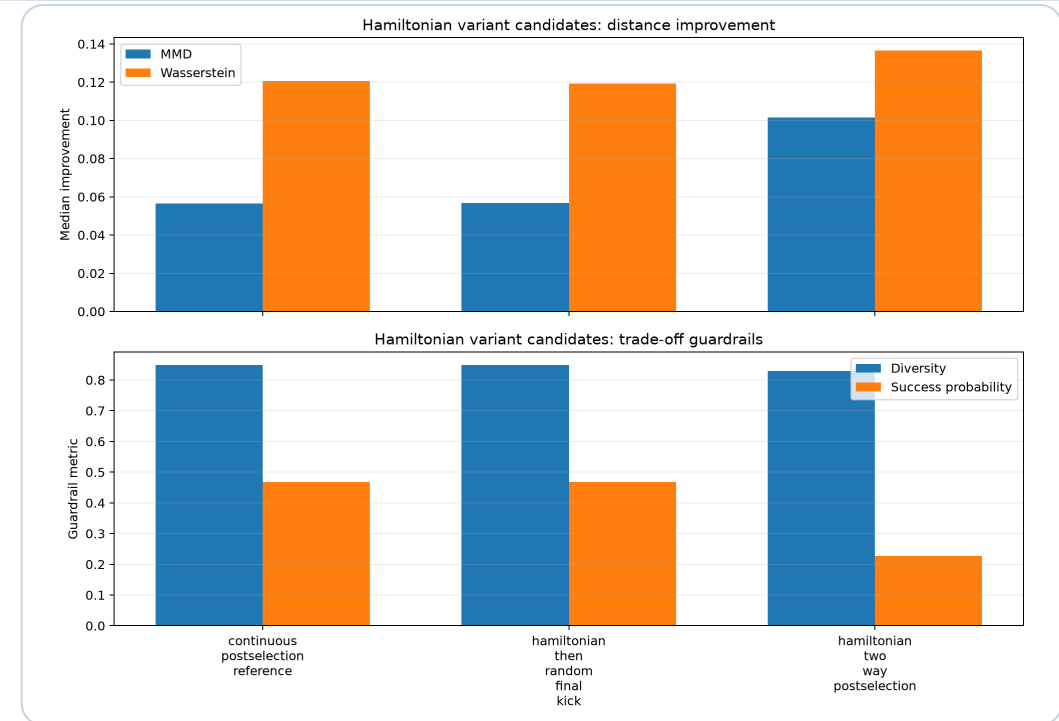


Figure: solution/figures/problem_3c_hamiltonian_variant_summary.png

Conclusion

Final thesis

Measurement-induced projection turns the complement-qubit basis into a control knob for an effective non-unitary denoising map, so Problem 3 should be judged by distance improvement, post-selection success, and diversity retention together.

- Novelty: the basis is a control knob, not just a scanned hyperparameter.
- Completeness: Problem 1(a) through Problem 3(c) are answered with traceable artifacts.
- Appropriateness: no full trainable QuDDPM or advantage claim; all claims stay within the small benchmark.
- Reproducibility: `source_code/` includes `src`, `scripts`, `tests`, `configs`, `submission/run_all.py`, and `solution`.

A1. Metric Definitions

- Fidelity: $F(\psi, \phi) = |\langle \psi | \phi \rangle|^2$.
- MMD uses a fidelity kernel between ensemble samples.
- Wasserstein-type distance uses pairwise cost $1 - F$.
- All Problem 1/2/3 comparisons use the same metric vocabulary.

A2. Hamiltonian and Projected Ensemble

- Start with data state $|\psi_i\rangle_M$ and complement state $|0\rangle_F$.
- Evolve $M+F$ by a fixed Hamiltonian: $|\Psi_i(t)\rangle = \exp(-iHt)(|\psi_i\rangle \otimes |0\rangle_F)$.
- Project F onto a measurement outcome basis vector and renormalize M .
- This is a projected diffusion benchmark, not a universal ranking over all Hamiltonian schedules.

A3. Measurement-Induced Non-Unitary Map

Conditional projected state

$$|\Psi_i(t)\rangle = \exp(-i H t)(|\psi_i\rangle_M \otimes |0\rangle_F)$$

$$|\phi_{i_m}(t, b)\rangle = (I_M \otimes \langle b_m |) |\Psi_i(t)\rangle / \sqrt{p_{i_m}}$$

$$p_{i_m} = \|(I_M \otimes \langle b_m |) |\Psi_i(t)\rangle\|^2$$

- Changing the basis b changes both the direction and strength of the effective map on M .
- Success probability is the operational cost proxy for post-selection.

A4. Axis-Only vs Continuous Explanation

Axis-only projection

Use Z/X/Y Pauli measurement bases. This is interpretable and easy to explain as a baseline.

Continuous post-selection

Search a continuous Bloch-sphere measurement basis. This generalizes axis-only control.

Important reading

- The margin over axis-only is small.
- The claim is not that continuous basis strongly or always wins.
- The useful result is that basis choice exposes a distance-success-diversity trade-off.

A5. Problem 3(c) Comparison Table

method	MMD	W	success	diversity
one-way ref	0.056388	0.120620	0.467554	0.848836
Ham+random	0.056695	0.119401	0.467554	0.848403
two-way Ham	0.101374	0.136426	0.227065	0.829273

- Two-way post-selection is the main 3-c candidate because it follows directly from the stronger-contraction trade-off seen in 3-b.
- It is not an unconditional win because success probability drops.

A6. Seed Robustness and Holdout Evidence

- 20 / 20 seeds pass the main Problem 3 adoption gate.
- Continuous 3(b) reference median gains: MMD 0.097056 and Wasserstein 0.147983.
- Guardrails: diversity retention 0.823217 and success probability 0.468122.
- The final notebook and tables preserve the trace from summary scripts to reported numbers.

A7. Source-Code Package and Reproduction Commands

Open `solution/solution_1.ipynb` first, then inspect `source_code/src` and `source_code/scripts`.

```
python -m pytest
```

```
python submission/run_all.py --quick
```

```
python scripts/summarize_problem_3_seed_sweep.py
```

```
python scripts/summarize_problem_3_method_portfolio.py
```

- `source_code/` includes `solution`, `src`, `scripts`, `tests`, `configs`, `submission`, `requirements.txt`, and `pyproject.toml`.
- `README_FOR_JUDGES.md` and `REPRODUCIBILITY_COMMANDS.md` provide the judge-facing map.

A8. Limitations and Guardrails

- Small 2-qubit/3-qubit state-vector benchmark.
- No quantum advantage, hardware advantage, or full trainable QuDDPM claim.
- Continuous basis margin over axis-only is small.
- Two-way post-selection improves distances but lowers success probability.
- Actor-critic, if mentioned, is target-aware and optional rather than a general unknown-target denoiser.
- The Haar baseline is a reference, not a training target.